

Christian Araya Soto

Data Analytics

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Hard Skills

- Data architecture and modeling (conceptual, logical, and physical data models).
- Implementation of relational (SQLite, MySQL), dimensional, and NoSQL databases.
- SQL databases for data ingestion, storage, and querying optimization.
- Data visualization, geospatial data, and interactive graphics.
- Development and calibration of predictive models (classification & regression).
- Experience with data strategies, business intelligence, and machine learning applications.
- Proficiency in R and Python for data analysis and modeling.
- ETL processes and data ingestion automation.
- Parallel programming in R (snow) and Python (joblib).
- Parallelization in the training and evaluation of machine learning models using scikit-learn.
- Machine Learning & AI: Decision Trees, Random Forest, Boosting, SVM, KNN.
- Unsupervised Learning: Hierarchical Clustering, k-means, PCA, UMAP, t-SNE.
- Deep Learning: Neural Networks, RNN, LSTM, optimization algorithms (Adam, RMSProp).
- Time Series Forecasting: ARIMA, Holt-Winters, recurrent neural networks, forecaster in R.
- Data Mining methodologies: CRISP-DM, KDD, regression models, model calibration.
- Dashboards & Visualization: R Shiny, Python Dash, Power BI.

Soft Skills

- Critical thinking and problem-solving.
- Team collaboration and leadership.
- Communication and data storytelling.
- Organizational capacity and strategic planning.
- Ability to work proactively and independently.

Work Experience

Mathematics Teacher (1995-2013)

Ministerio de Educación Pública, Costa Rica

- Extended learning experiences beyond the classroom.
- Communicated with parents and counselors regarding student progress.
- Prepared students for national tests; maintained high ethical standards.
- Facilitated learning using instructional methods and technology to enhance critical thinking.

Administrator (2000-2024)

Centro de Apoyo Educativo

- Founded and managed company operations, staff recruitment, payroll, and scheduling.
- Oversaw accounts receivable, advertising, and quality standards.
- Designed study plans, implemented online tutoring, and coordinated academic programs.

Education

- Universidad Estatal a Distancia – Present | Computer Engineering
- Promidat – Present | Executive Master in Data Science & AI - Parallel programming in R and Python - Linear algebra, calculus and optimization applied to AI - Advanced probability and statistics - Supervised and unsupervised methods in R and Python - Advanced dashboard development - Final research project (Thesis)
- Promidat – 2024 | Expert Program in Data Mining with R - Exploratory, predictive, and regression methods - Clustering, dimensionality reduction, PCA - Penalized regression (LASSO, Ridge) - Time series forecasting (ARIMA, Holt-Winters, RNN, LSTM) - Cross-validation and model selection
- American University – 2019 | Bachelor's in Mathematics Teaching
- Costa Rica University – 1998 | 6th Semester Civil Engineering

Certifications & Credly Badges

- Executive Master in Data Science & AI (Promidat - 2025)
- Expert in Data Mining with R (Promidat - 2024)
- Cisco CCNA 1 (2013)

Languages

Spanish (Native)

English (B2)